Massey University
College of Sciences

161.304: Statistical Modelling

Assignment 2, 2024

GENERAL INFORMATION: This assignment is assessed. It carries weight 25% towards your final mark for the 161.304 paper. Work should be submitted electronically via the Stream site for 161.304, preferably as a PDF document with all your works, including your answer, R code, and any references used.

DEADLINE: Your work for this assignment must be submitted by **11.55pm on Friday 11 October**. Late penalty - 0 mark unless prior arrangement.

PLAGIARISM: The work that you submit must be your sole effort (i.e. not copied from anyone else). If you are found guilty of plagiarism you will be penalised. Happy for you to use ChatGPT to improve your code or your English, only state how you have used it if you have. It is an amazing tool and much can be learned form its use. Automatically generated solutions are still easy to spot and often weird, so please do not try to use this tool without thinking.

EXERCISE 1: Spurious association

1. Consider the "Overthinking box: Simulating spurious association" in Section 5.1 of the *Statistical Rethinking* book. Finish the analysis to -if possible- find the true DAG of the system using relevant regression analysis technique(s).
[6 marks]

2. Similarly, consider the "Overthinking box: Simulating a masking relationship" in Section 5.2 of the *Statistical Rethinking* book. Finish the analysis to -if possible- exhibit the true DAG of the system using relevant analysis techniques.
[6 marks]

EXERCISE 2: Another DAG for the Marriage example
Do exercises 5H1 and 5H2 of Chapter 5 of the *Statistical Rethinking* book.
[16 marks]

EXERCISE 3: Foxy DAG and models
Do exercises 6H3, 6H4 and 6H5 of Chapter 6 of the *Statistical Rethinking* book.
[18 marks]

Then do exercise 7H5 of Chapter 7.
[10 marks]

EXERCISE 4: Markov Chains

1. A Markov chain with state space $S = \{1, 2\}$ has transition matrix

$$P = \begin{bmatrix} 0 & 1 \\ 1-\delta & \delta \end{bmatrix}$$

where $0 < \delta < 1$. Find the stationary distribution for this process (as a probability vector with elements depending on δ). Show your full working.
[4 marks]

2. Find the stationary distribution for the Markov chain with state space $S = \{1, 2, 3, 4\}$ and transition matrix

$$P = \begin{bmatrix} 0.2 & 0.4 & 0.2 & 0.2 \\ 0.1 & 0.6 & 0.2 & 0.1 \\ 0.01 & 0.98 & 0 & 0.01 \\ 0.4 & 0.3 & 0.2 & 0.1 \end{bmatrix}.$$

Provide details of your working either as mathematical calculations or R code.

[3 marks]

EXERCISE 5: Are you warm enough?

Do exercises 9M3 of Chapter 9 of the *Statistical Rethinking* book.

[10 marks]

EXERCISE 6: Fitting models with a MCMC algorithm

Do exercise 9H2 of Chapter 9 and compare the results to the ones obtained with the quadratic approximation (`quap()` fitting function).

[12 marks]